

Language Identification from Audiobook Speech: A Technical Report

1 Introduction to Language Identification

1.1 Definition and Importance

Spoken Language Identification (LID) is the automated detection of language from speech segments. Applications include:

- **Voice assistants:** Automatic language switching (Alexa, Siri)
- **Multilingual systems:** Content routing and processing
- **Call centers:** Language-based query distribution
- **Media monitoring:** Broadcast content classification

1.2 Controlled vs. Real-world Data

- **Controlled recordings:** Studio environments with consistent conditions. (e.g., Common Voice dataset) Sudden changes in tone or noise are very rare
- **Real-world audiobooks:** Variable quality, emotional expression, narrative pacing. The model should be robust to tone and detect language from words.

1.3 LID Paradigms

Closed-set LID: Identifies among known languages only.

- *Implementation:* Multi-class classification.
- *Use case:* All possible languages are predefined. for example in this project, there are a handful of available languages. or an app where it has implementation for only five language.

Open-set LID: Identifies or rejects unknown languages.

- *Implementation:* Binary classifiers per language

- *Use case:* Unknown languages may appear, model can determine which group of language has the most unknown rate and for example, stakeholders can determine which language to add next with more insight.

2 Challenges in Language Identification

2.1 Key Challenges

1. **Similarity:** Italian, Spanish and French are all latin based. and have high lexical similarity.
2. **Accent:** Regional differences (Roman vs. Milanese Italian)
3. **Noise:** Spoken language has high noise in urban cities
4. **Device effects:** Recording quality vary greatly.
5. **Short segments:** If a segment is too short, model performance may be low or at best inconsistent.

2.2 Solution Strategies

- **Data collection:** Get data from every source (both high and low quality, different accent and so on) so that the model know not to overfit on an accent or a single pronunciation
- **Data augmentation:** Speed perturbation, noise injection so reduce noise effect in classification.
- **Overfit solutions:** By introducing tools like regularization or drop out layer, the model will be forced to pay attention to all features and not classify by just a single feature.

3 Audio Data Characteristics and Selection

3.1 Selection Criteria for LID

- As discussed in previous section, one way to overcome diversity in spoken language is to have diverse training data. if we have data from all scenarios it will be good. but it is not necessary. for example, if we want to automatically detect language of an audiobook, noise will play a much smaller effect in data.
- An ideal data is one that is as diverse as we want to and has a suitable segment size so that the model can actually learn pattern in voices.
- tone, speech, amplitude, phonetic and pronunciation are some of the features where humans use to detect language.

3.2 Desirable Properties

1. Clean speech with minimal background noise or music
2. Multiple speakers for gender and age diversity
3. Balanced phoneme representation
4. Optimal segment duration (long enough so that model can learn pattern but not too long.)

3.3 Qualitative Evaluation

Listening assessment revealed:

- Clear articulation in audiobook speech
- Consistent recording quality
- Emotional variation within narrative
- Potential bias toward literary vocabulary
- Bias can be incurred from reader (high speed reader, gender, etc)

4 Feature Extraction Techniques

Table 1: Feature Extraction Methods Comparison

Feature	How it Works	LID Suitability	Advantages	Limitations
MFCC	Models human auditory perception via mel filterbanks	High	Robust to noise, standard in speech processing	Loses spectral detail
FFT	Frequency domain representation	Low	Complete frequency information	Sensitive to noise
Log-Mel Spectrogram	Power spectrum on mel scale	Medium	Preserves temporal patterns	High dimensionality
Spectral Centroid	Center of mass of spectrum	Low	Simple to compute	Language-insensitive
Chroma Features	Maps to 12 pitch classes	Very Low	Pitch-invariant	Designed for music
Spectral Contrast	Difference between peak/valley	Medium	Robust to loudness variations	Computationally expensive
Zero-Crossing Rate	Rate of sign changes	Low	Simple, fast	Limited discriminative power
LPC	Linear prediction coefficients	High	Compact representation, efficient	Assumes source-filter model
PLP	Perceptually weighted LPC	High	Noise robust, compact	Complex computation

5 Similarity Learning for Language Discrimination

5.1 Definition

Similarity Learning learns a metric space where samples of the same language are closer than samples of different languages, without explicit classification boundaries. Sometimes

its called contrastive learning.

5.2 Loss Functions [?]

1. Contrastive Loss:

$$L = (1 - Y) \cdot D^2 + Y \cdot \max(0, m - D)^2$$

Where D is distance, $Y = 1$ for same language, m is margin

Contrastive loss is used to minimize the distance between instances of same group.

2. Triplet Loss:

$$L = \max(0, d(a, p) - d(a, n) + \text{margin})$$

Requires triplets: anchor a , positive p (same language), negative n (different)

It calculate whether a is near p or q .

5.3 Application to LID

We can check the new data is near to instances of which class and then classify it (something like KNN).

In order to do this, we need to get a good feature extraction where instances of different classes are far from each other and instances of same class are near together. so that whatever loss we use, its value for pairs of same class must be lower than pairs of different classes.

Assuming we have extracted (or generated) such a feature, then we can calculate contrastive or triplet loss and do the classification by returning nearest instance(or k nearest instances) class.

5.4 Advantages

- Enables open-set recognition
- Learns language-specific subspaces
- Robust to unseen accents/variants
- Usually needs lower data than a full classification (but not always)
- Although it is supervised learning, it can be used to discover new unknown unlabeled data because it partitions the feature space and we can infer that this new language is like french but with a hint from arabic (just a wild undiscovered language!)

5.5 Implementation Strategy

As discussed in feature extraction section, MFCC features are suitable for LID. we can extract MFCC features (along with some other useful feature like LPC) then using a Siamese NN [5] we calculate distance from samples and then classify it by something like KNN

6 References

References

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